# Complete Paper Solution

# EGD – Winter 2023

## Engineering Graphics and Design

## (GTU)

**Q.1 (a)** Define Representative Fraction. With reference to Representative Fraction, classify the scales.**( 03 Marks)**

**Answer:**

Representative Fraction is the ratio of size of object in paper to actual size of object.

$$\text{Representative Fraction (R.F.)} = \frac{\text{Size of object in paper}}{\text{Actual size of object}}$$

According to value of R.F., we can explain full scale, reduced scale and enlarged scale.

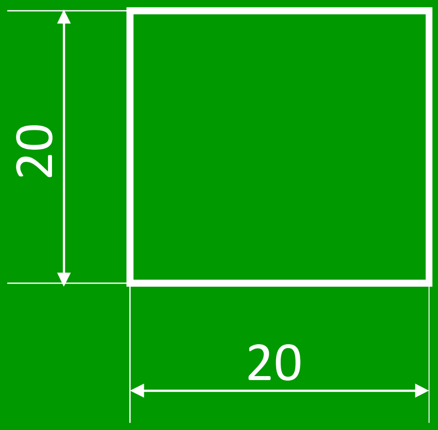
**Full Scale:**

Draw 20mm size square.

Representative Fraction (R.F.) = $\dfrac{\text{Size of object in paper}}{\text{Actual size of object}}$

Representative Fraction (R.F.) = $\dfrac{20mm}{20mm} = \dfrac{1}{1}$ = 1:1

Here, size of object in paper and actual size of object is same.




Scale 1:1

**Use:**

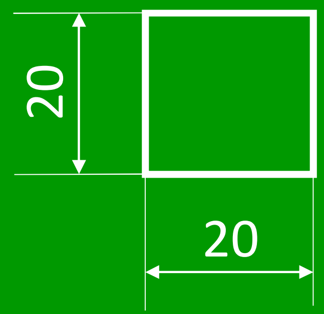
It is used in general purpose drawing.

## Reduced Scale:

Draw 20mm size square.

$$\text{Representative Fraction (R.F.)} = \frac{\text{Size of object in paper}}{\text{Actual size of object}}$$

$$\text{Representative Fraction (R.F.)} = \frac{10\text{mm}}{20\text{mm}} = \frac{1}{2} = 1:2$$




Scale 1:2

Here, size of object in paper is less than actual size of object.

## Other Example:

1:5, 1:10, 1:20, 1:50, 1:100, 1:1000, 1:100000

## Use:

When size of object is too large.

Example: Building drawing, Machinery drawing, map etc.

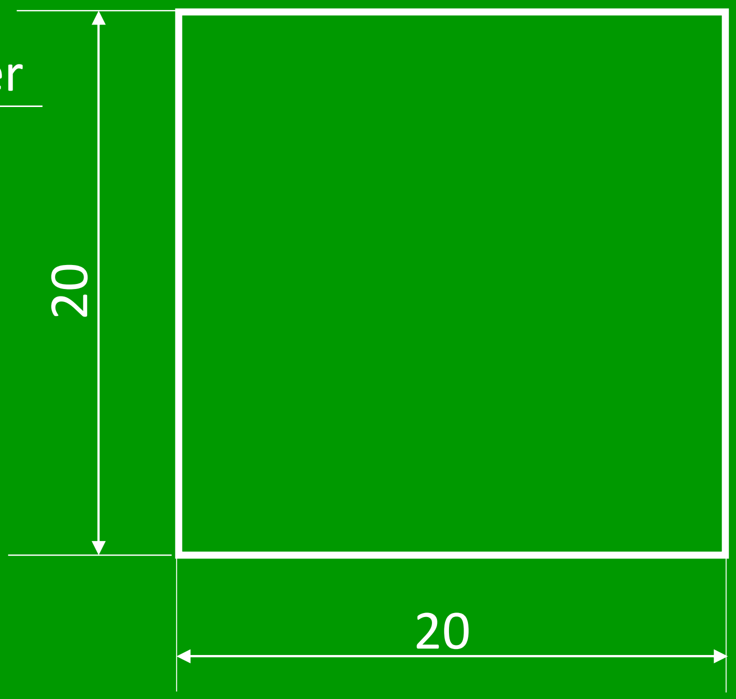
**Enlarged Scale:**

Draw 20mm size square.

$$\text{Representative Fraction (R.F.)} = \frac{\text{Size of object in paper}}{\text{Actual size of object}}$$

$$\text{Representative Fraction (R.F.)} = \frac{40\text{mm}}{20\text{mm}} = \frac{2}{1} = 2:1$$

Here, size of object in paper is greater than actual size of object.




Scale 2:1

**Other Example:**

5:1, 10:1, 20:1, 50:1 etc.

**Use:**

When size of object is too small.

Example: Parts of wrist watch, parts of electronic circuit etc.

**Q. 1 (b)** Construct a scale of 1:40 to read meters and decimeters and long enough to measure 6 m. Mark on it a distance of 4.7 m and 3.2 m. **(04 Marks)**

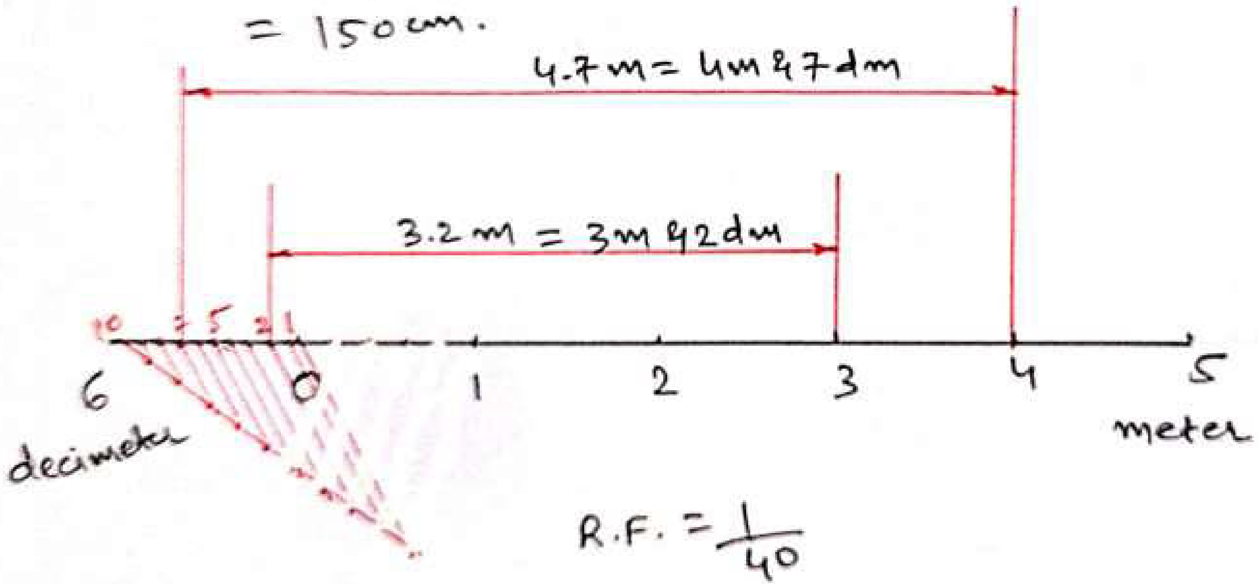
1) $R.F. = \frac{1}{40}$

2) $A.L.S. = R.F. \times Max.\ Measurement$

$= \frac{1}{40} \times 6m$

$= \frac{1}{40} \times 6 \times 100\ cm$

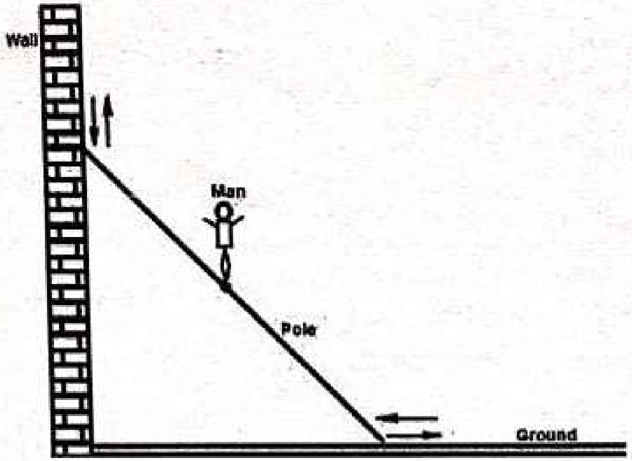
$= 150\ cm.$




1m = 10dm

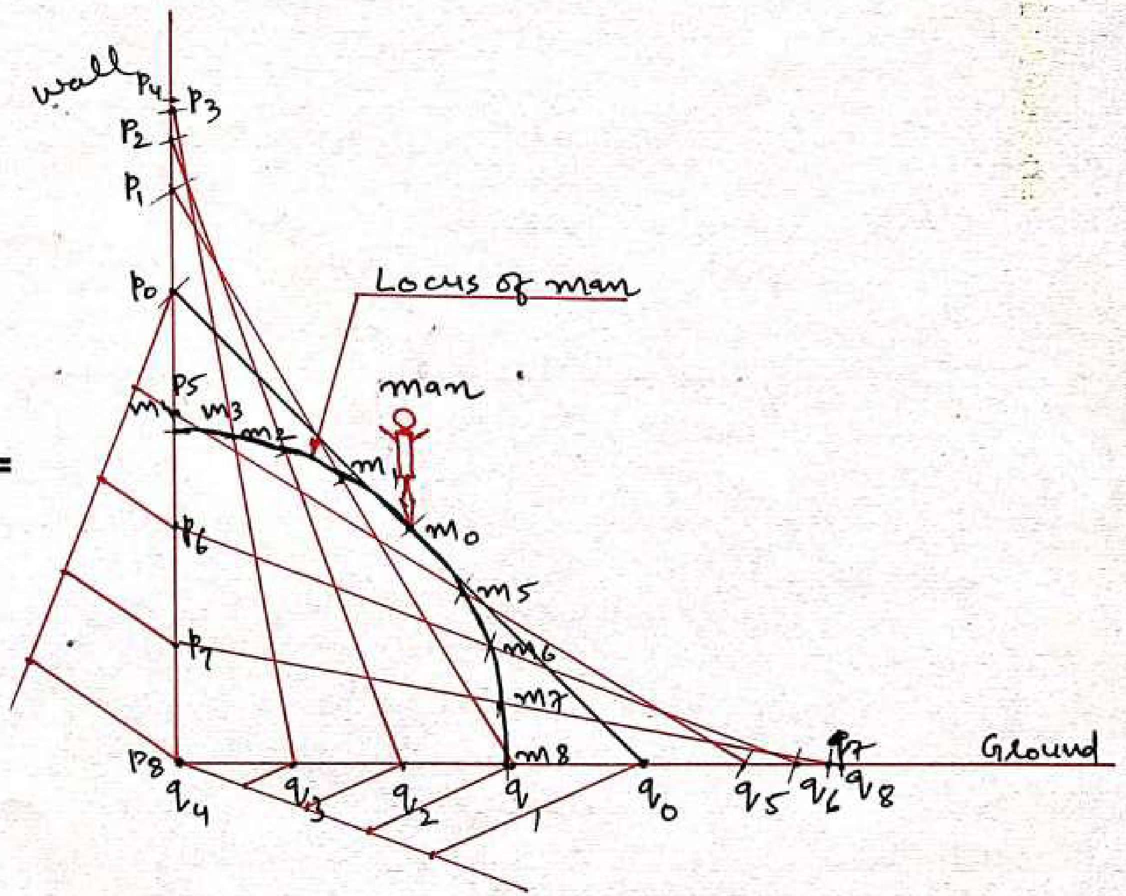
$R.F. = \frac{1}{40}$

**Q. 1 (c)** A pole is sliding vertical to horizontal position as shown in figure. A man is standing at the midpoint of pole. Trace locus of man for sliding of the pole from vertical to horizontal. Length of pole is 480 centimeters. **(07 Marks)**

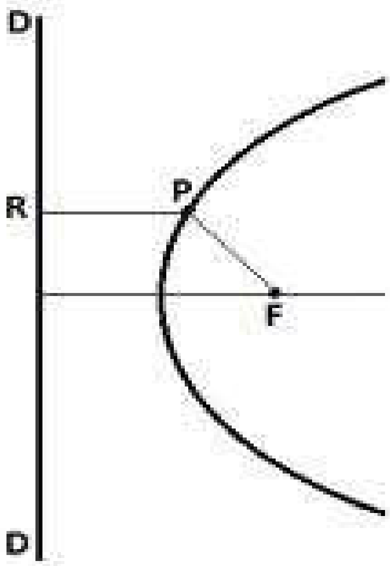



Scale

480 cm = 10 cm



**Q.2 (a)** In the figure shown here, show the conditions for having Ellipse, Parabola and Hyperbola for the following equation PF/PR (03 Marks)

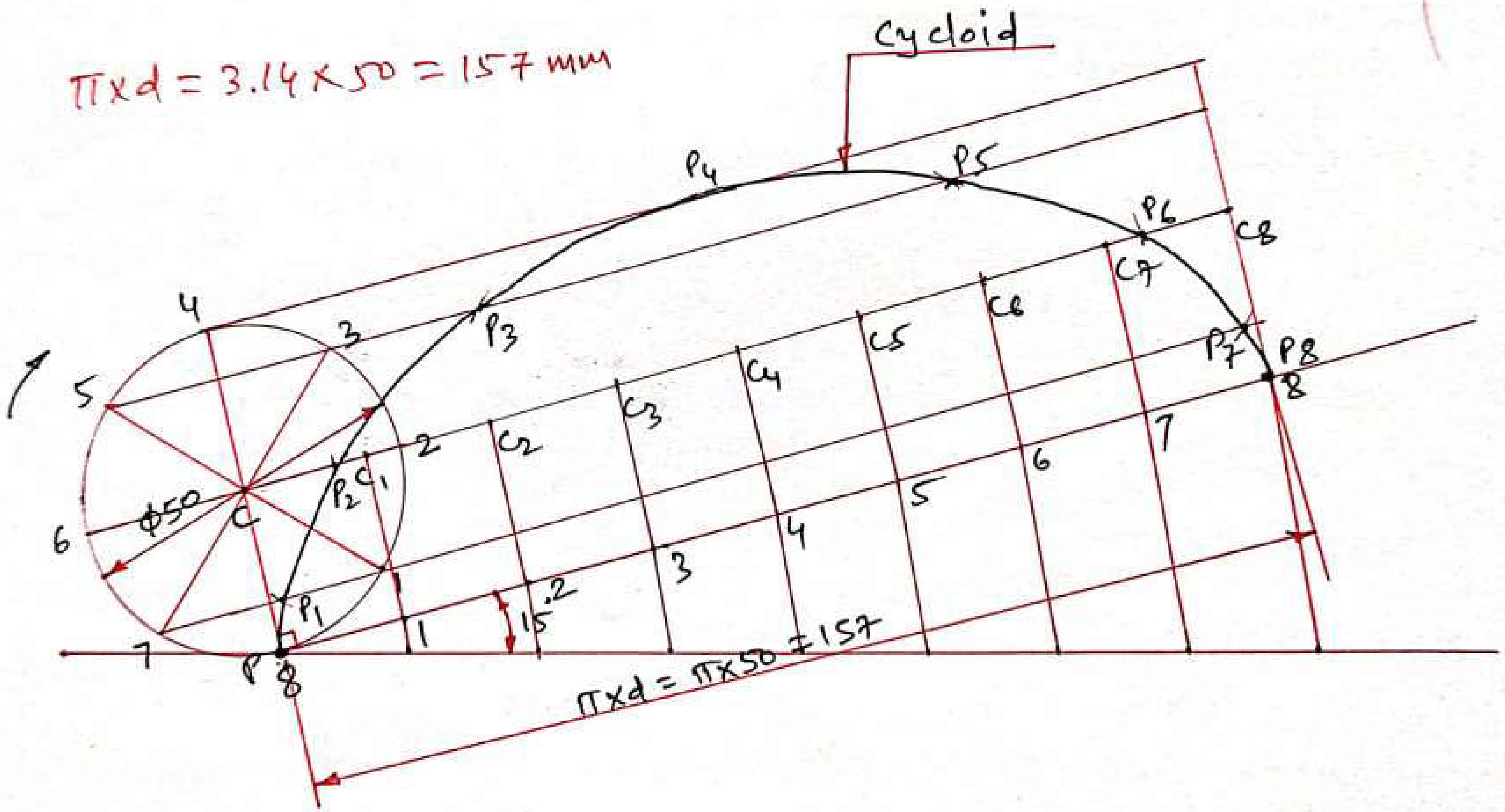



- In this figure the ratio PF/PR (Distance between PF / Distance between PR) is known as **eccentricity.**
- So, if the value of this ratio PF/PR < 1 then the curve obtained is **Ellipse**.
- If value of this ratio PF/PR =1 then the curve obtained is **Parabola**.
- If value of this ratio PF/PR > 1 then the curve obtained is **Hyperbola**.

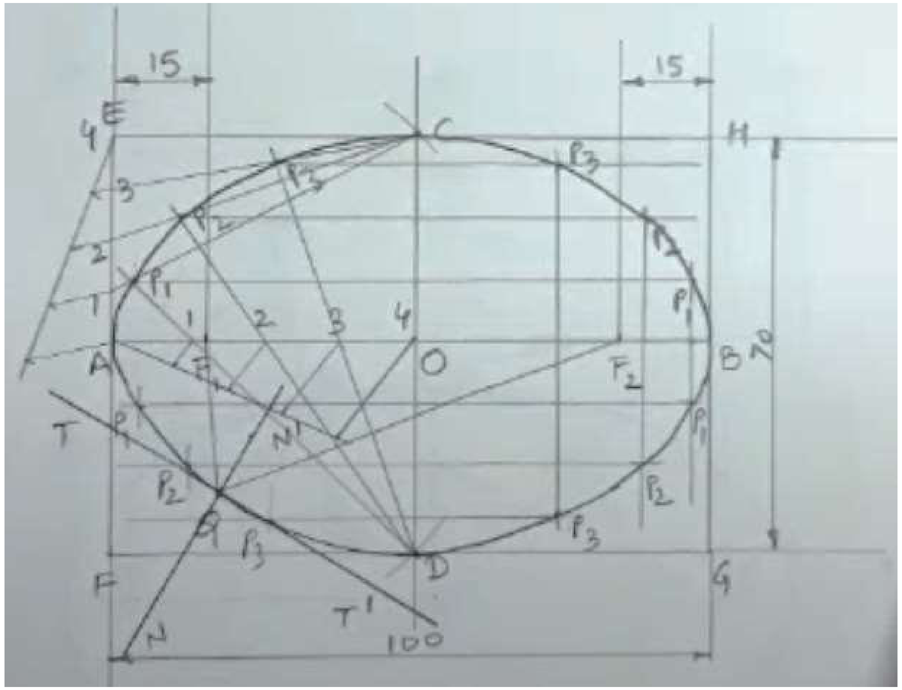
**Q.2 (b)** Identify the following types of line and write their use in engineering drawing.
(04 Marks)

| | **Types of Line** | **Use** |
|---|---|---|
| 1)  | Short dash medium | To show hidden outlines and edges in the object |
| 2)  | Long chain thin | To show centre line, locus line, pitch line and pitch circle |
| 3)  | Long chain thick at ends and thin else where | To show cutting plane in the object |
| 4)  | Short zig zag thin | To show long break lines |

**Q. 2 (c)** A thin circular disc of 50 mm diameter is allowed to roll without slipping from upper edge of sloping board which is inclined at 15° with the horizontal plane. Draw the curve traced by the point on the circumference of the disc. Consider the point is at contact surface of disc and sloping board initially. **(07 Marks)**

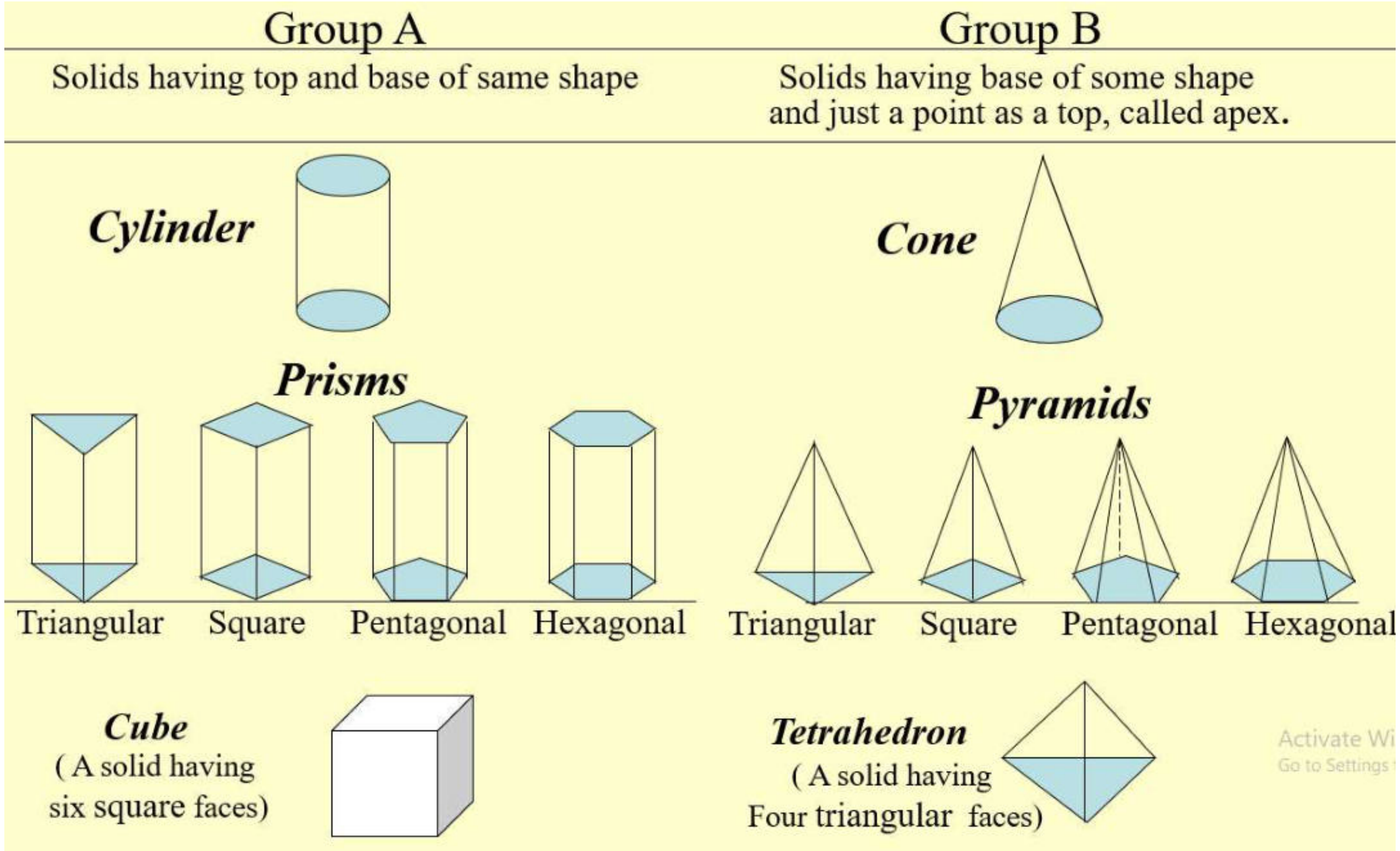


**Q.2 (c) OR** Major axis of an ellipse is 120 mm and the distance between focii of ellipse is 80 mm. Draw an ellipse using oblong method. Draw tangent and normal to any point to the curve. **(07 Marks)**



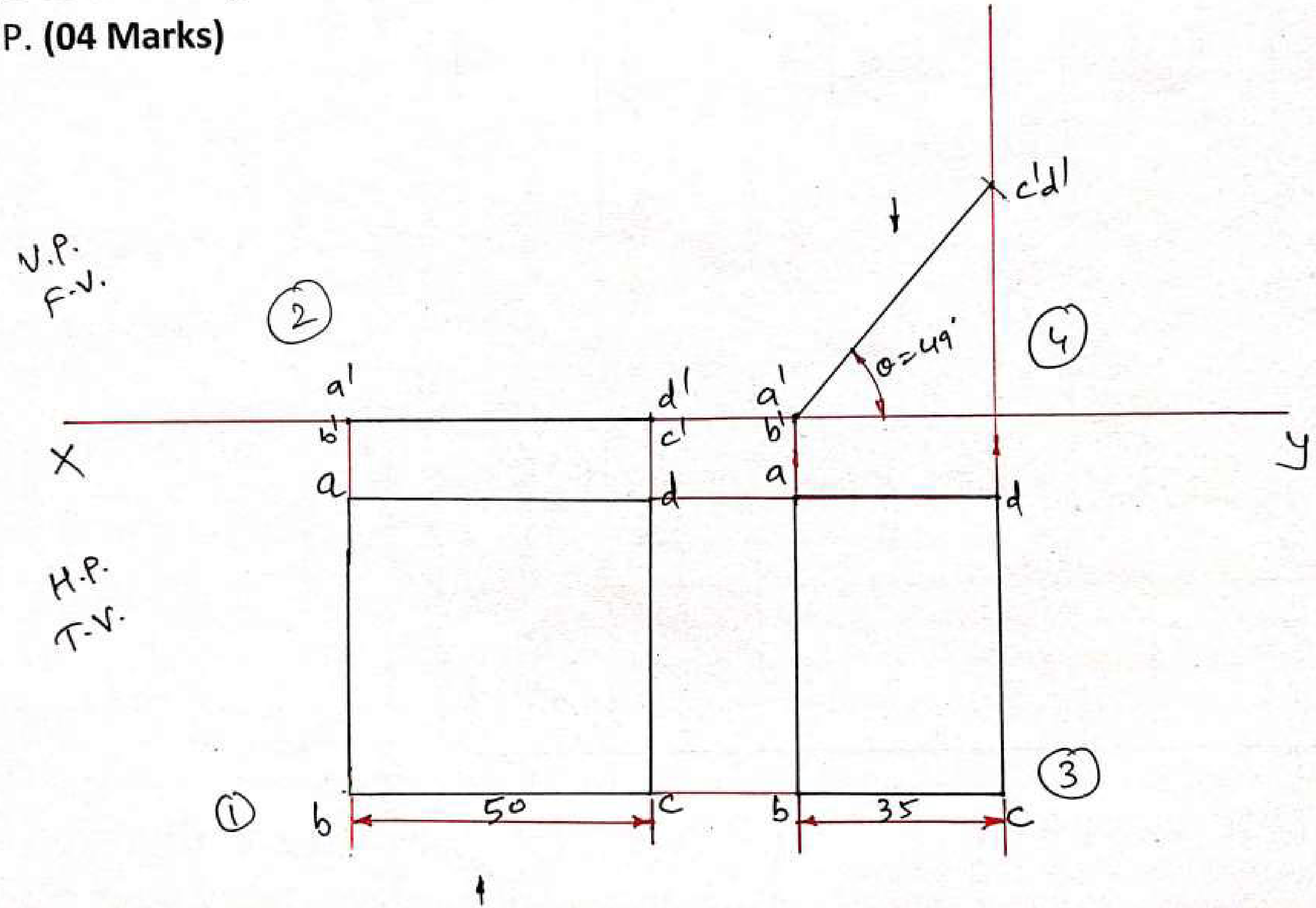

[Video Link: https://youtu.be/Rcw2kGysMw4](https://youtu.be/Rcw2kGysMw4)

**Q.3 (a)** Differentiate between prism and pyramid. **( 03 Marks)**

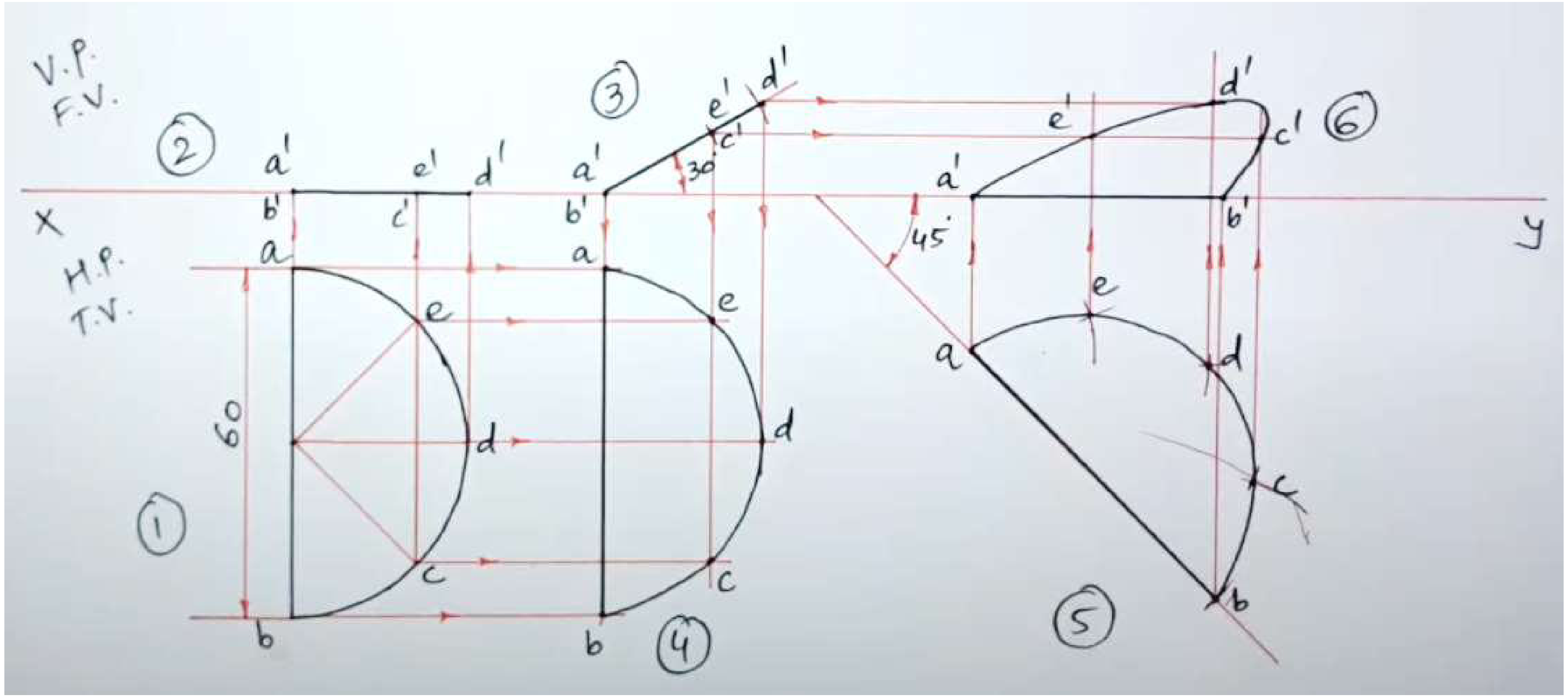



[Video Link: https://youtu.be/8pFcScIZ1Hs](https://youtu.be/8pFcScIZ1Hs)

**Q. 3 (b)** A square ABCD of 50 mm side is resting on HP on one of its side such that it appears as a rectangle of 35 mm X 50 mm. Draw its projection and find its inclination with HP. **(04 Marks)**

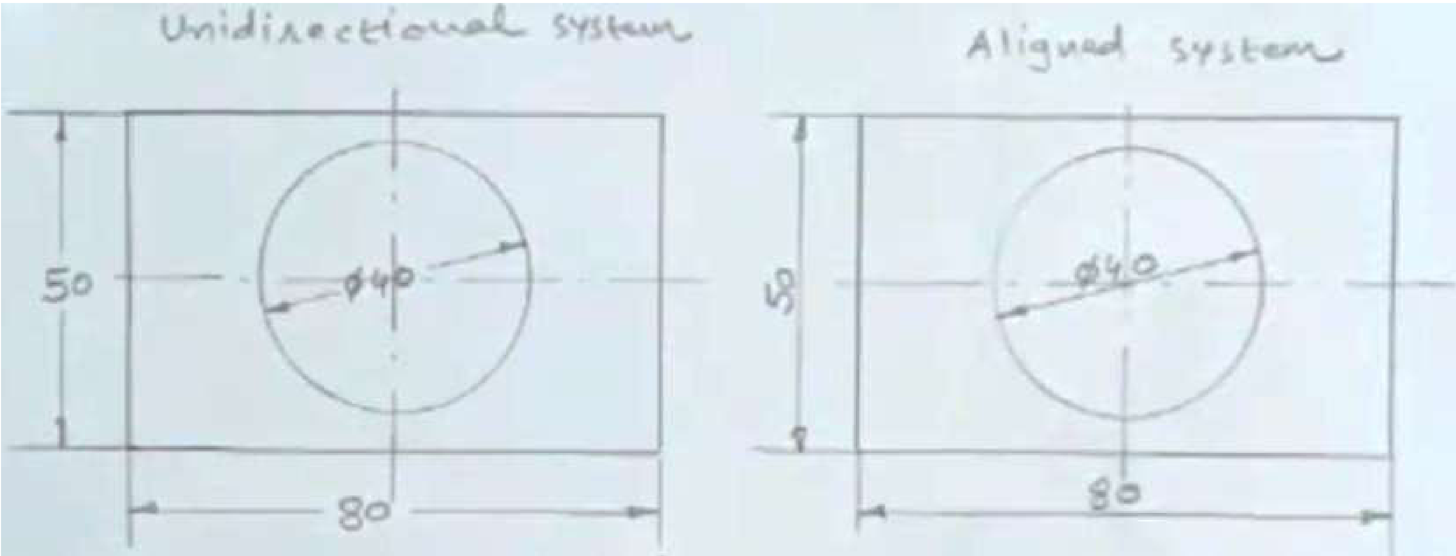


**Q.3 (c)** A semicircular plate at has the 60 mm diameter straight edge in HP and inclined at 45o to VP. The surface of plate makes an angle of 30o to HP. Draw its projection.
**(07 Marks)**



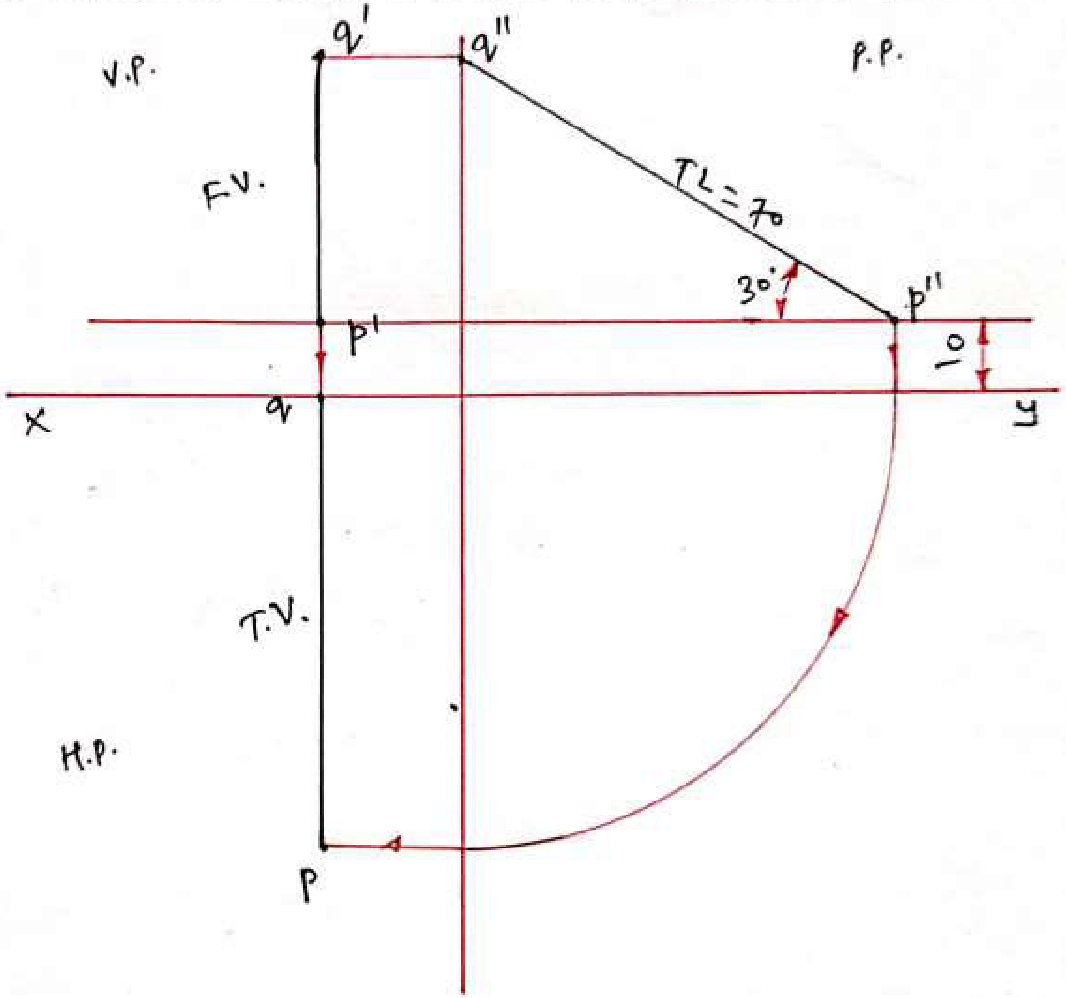

Video Link : https://youtu.be/EyNcH5LDZNg

**Q.3 (a) OR** Show by sketches the difference between (i) aligned dimensioning and (ii) unidirectional dimensioning **(03 Marks)**




Video Link : https://youtu.be/sGyWLKiFhKM

**Q.3 (b) OR** A line PQ 70 mm long is parallel to profile plane and inclined at 30° to HP. Point P is 10mm Above HP and point Q is in VP. Draw its projections. **(04 Marks)**



**Q. 3 (c) OR** A line AB, 80mm long is inclined to HP at 30° and to VP at 45°. The end A is in HP and VP. Draw projection of line AB. Find length of top view and front view of line and inclination of them with HP and VP. **(07 Marks)**

T.L. = 80mm

$\theta = 30^\circ$

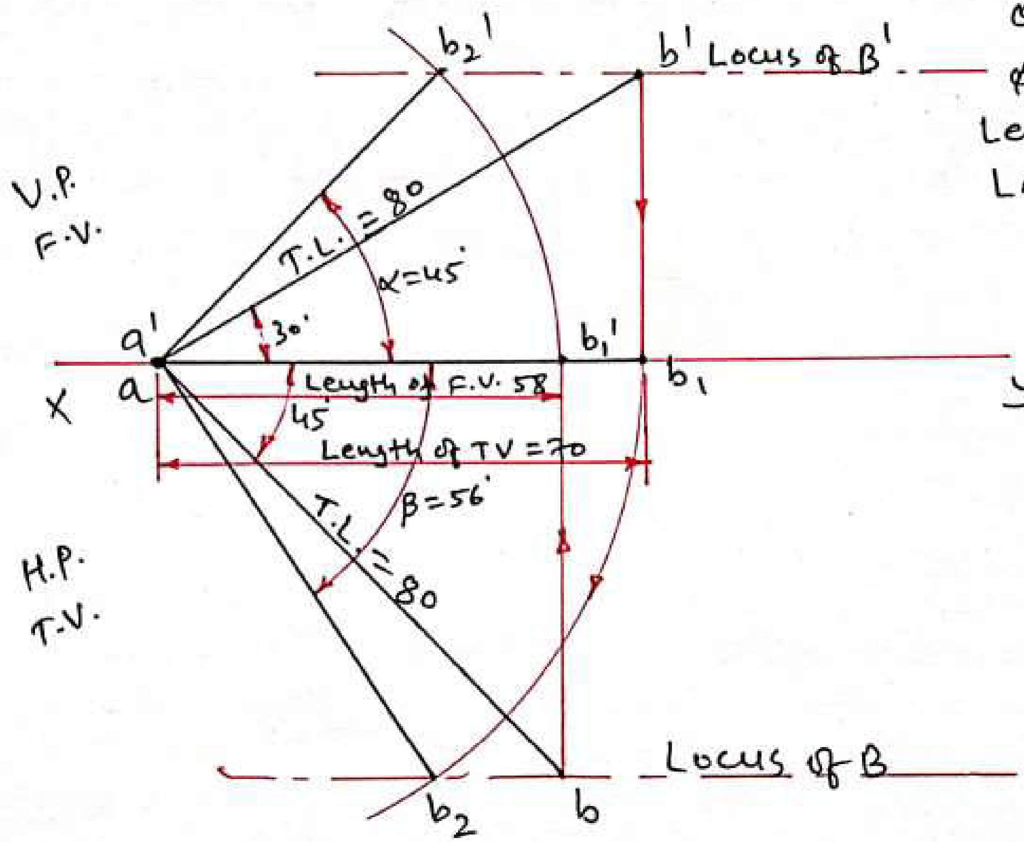
$\phi = 45^\circ$

Length of T.V. = ? 70 mm

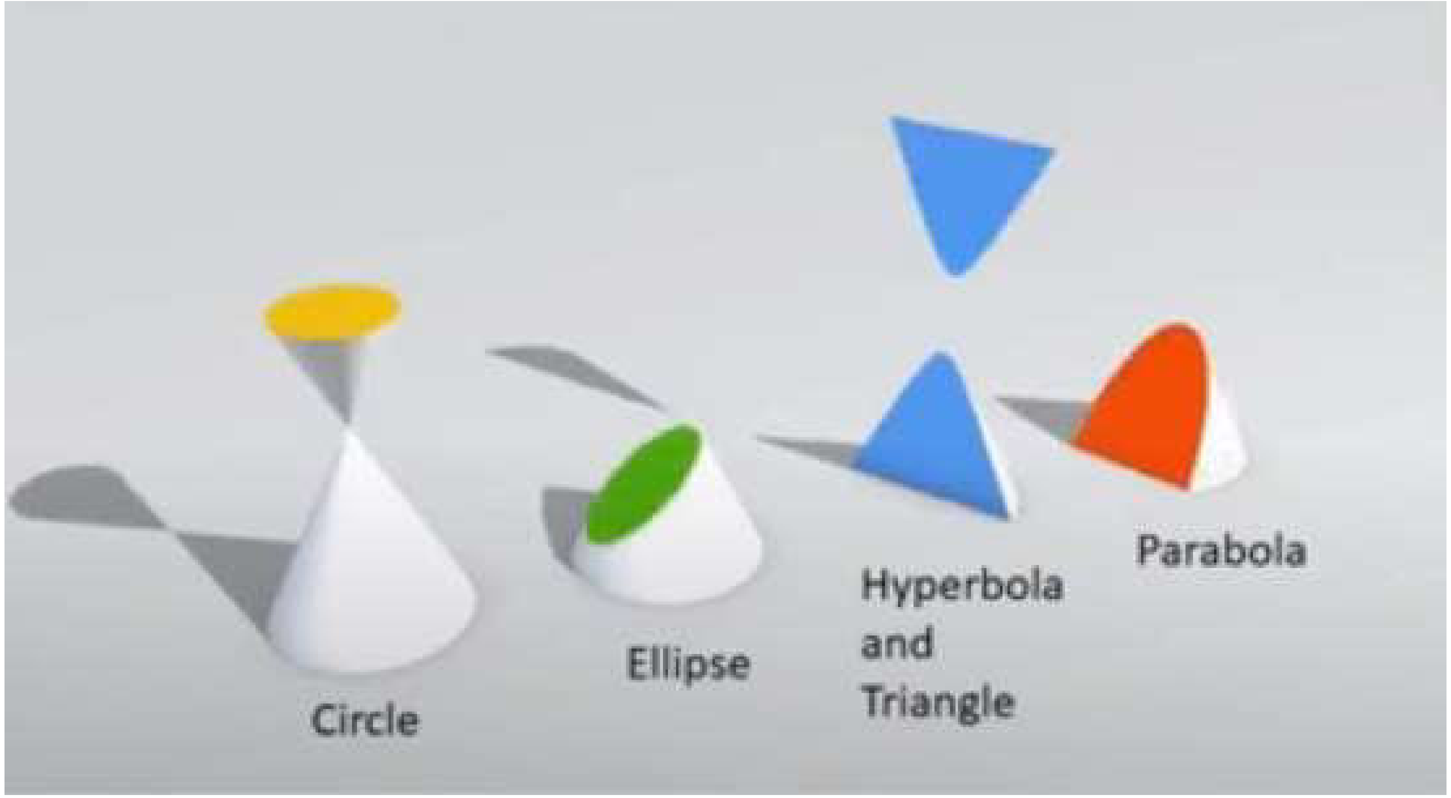
Length of F.V. = ?

$\beta$ = ? $56^\circ$

$\alpha$ = ? $45^\circ$



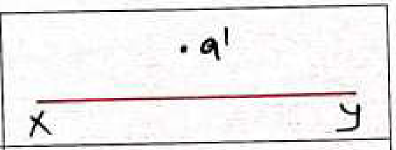
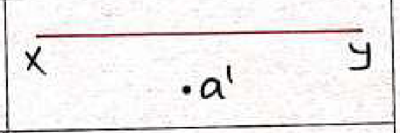
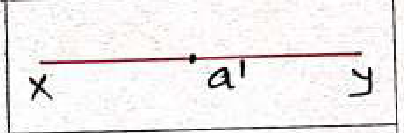
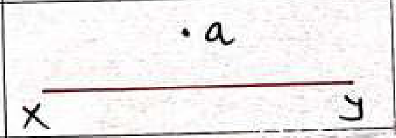
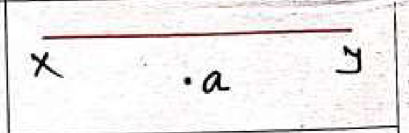
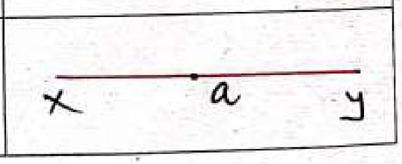
**Q.4 (a)** Define Parabola, Ellipse and Hyperbola **(03 Marks)**

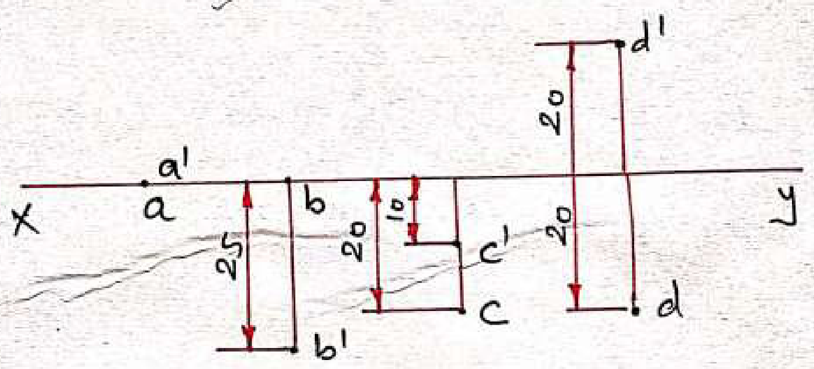



Video Link https://youtu.be/oZrHC_o5Klw

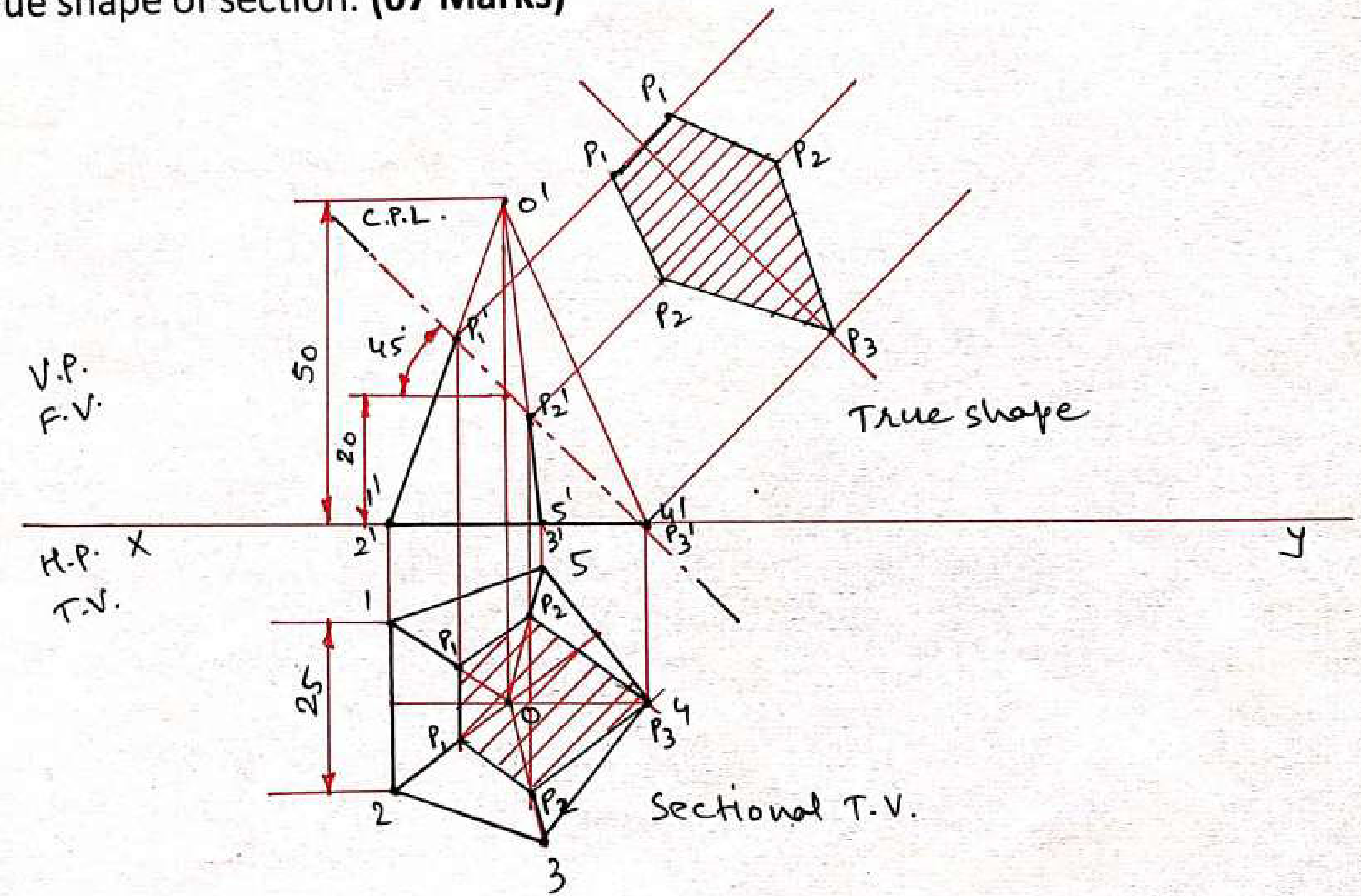
**Q.4 (b)** Draw projection of following points:

1. Point A is in HP and in VP
2. Point B is in VP and 25 mm below HP
3. Point C is 10 mm below HP and 20 mm in front of VP
4. Point D is 20 mm above HP and 20 mm in front of VP **(04 Marks)**

| | |
|---|---|
| Point A above H.P. |  |
| Point A below H.P. |  |
| Point A on/in the H.P. |  |
| Point A behind V.P. |  |
| Point A in front of V.P. |  |
| Point A on/in the V.P. |  |



**Q. 4 (c)** A pentagonal pyramid, side of base 25 mm and height 50 mm is resting on HP with one of the edge of base is perpendicular to VP. It is cut by auxiliary inclined plane inclined at 45° and cuts the axis at 20mm above the base. Draw sectional top view, front view and true shape of section. **(07 Marks)**

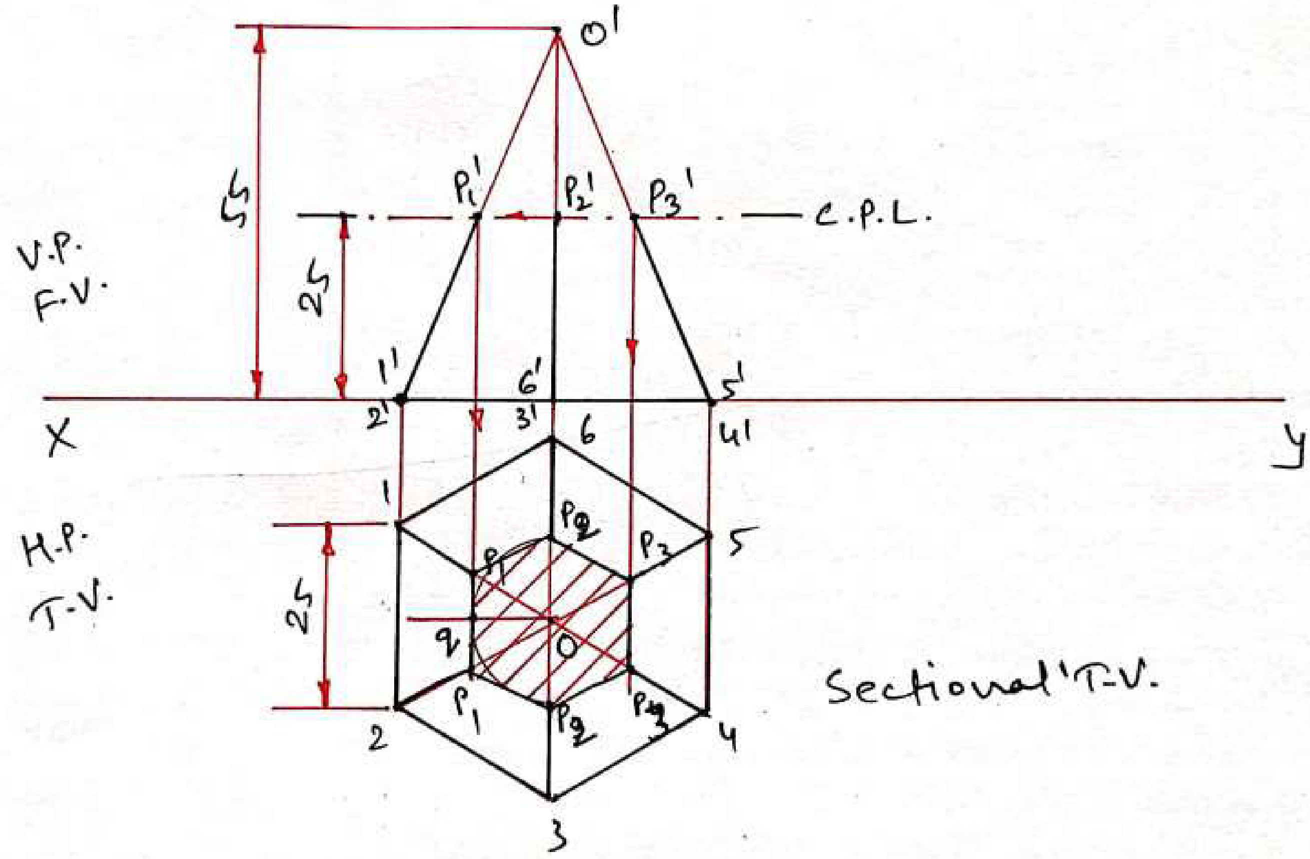


**Q.4 (a) OR** Benefits of AUTOCAD over conventional drawing techniques **( 03 Marks)**

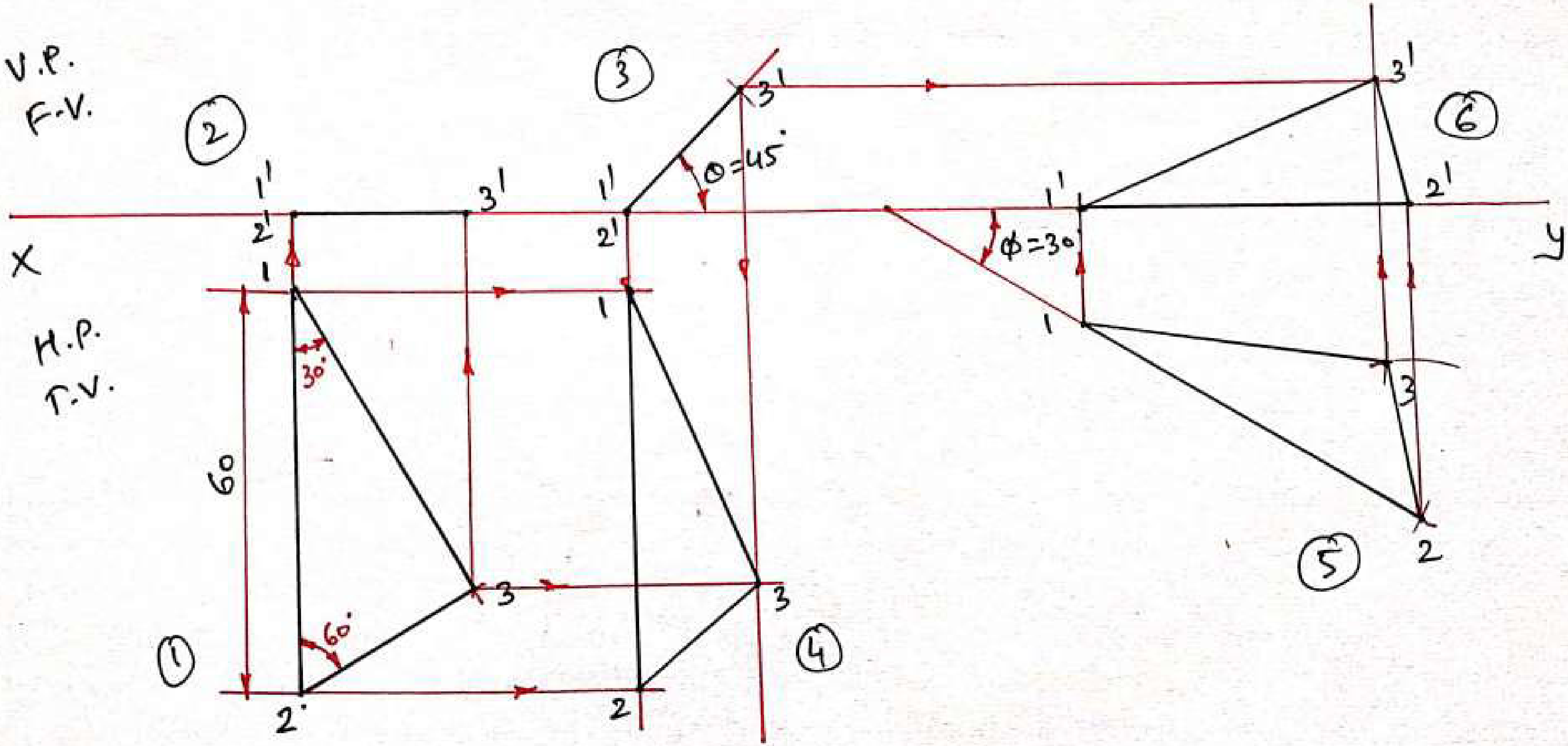
Benefits of AutoCAD is as per the following.

1. With the help of CAD highly accurate and precise drawing can be prepared in less time than manually prepared drawings.
2. Drawing instruments and hand skill is not required.
3. Drawings can be easily saved in computer and it can be modified and copied when necessary.
4. We can print the drawings on printer and plotter when required.
5. No scrap of drawings and no paper wastage.
6. Multicolor display makes visualization more effective.
7. We can increase of decrease the size of drawing to definite scale very easily.
8. Computer aided manufacturing is also possible.

**Q. 4 (b) OR** Hexagonal pyramid of side of base 25 mm and height 55mm is resting on HP on its base. It is cut by a plane parallel to HP and perpendicular to VP and cutting the axis at 25 mm above the base. Draw its projection. **(04 Marks)**



**Q.4 (c) OR** A 30° – 60° set square of 60 mm longest side is so kept such that the longest side is in HP making an angle of 30° with VP. The set square itself is inclined at 45° to HP. Draw the projections of the set square. **(07 Marks)**



**Q.5 (a)** What are the advantages and disadvantages of using AUTOCAD? **( 03 Marks)**

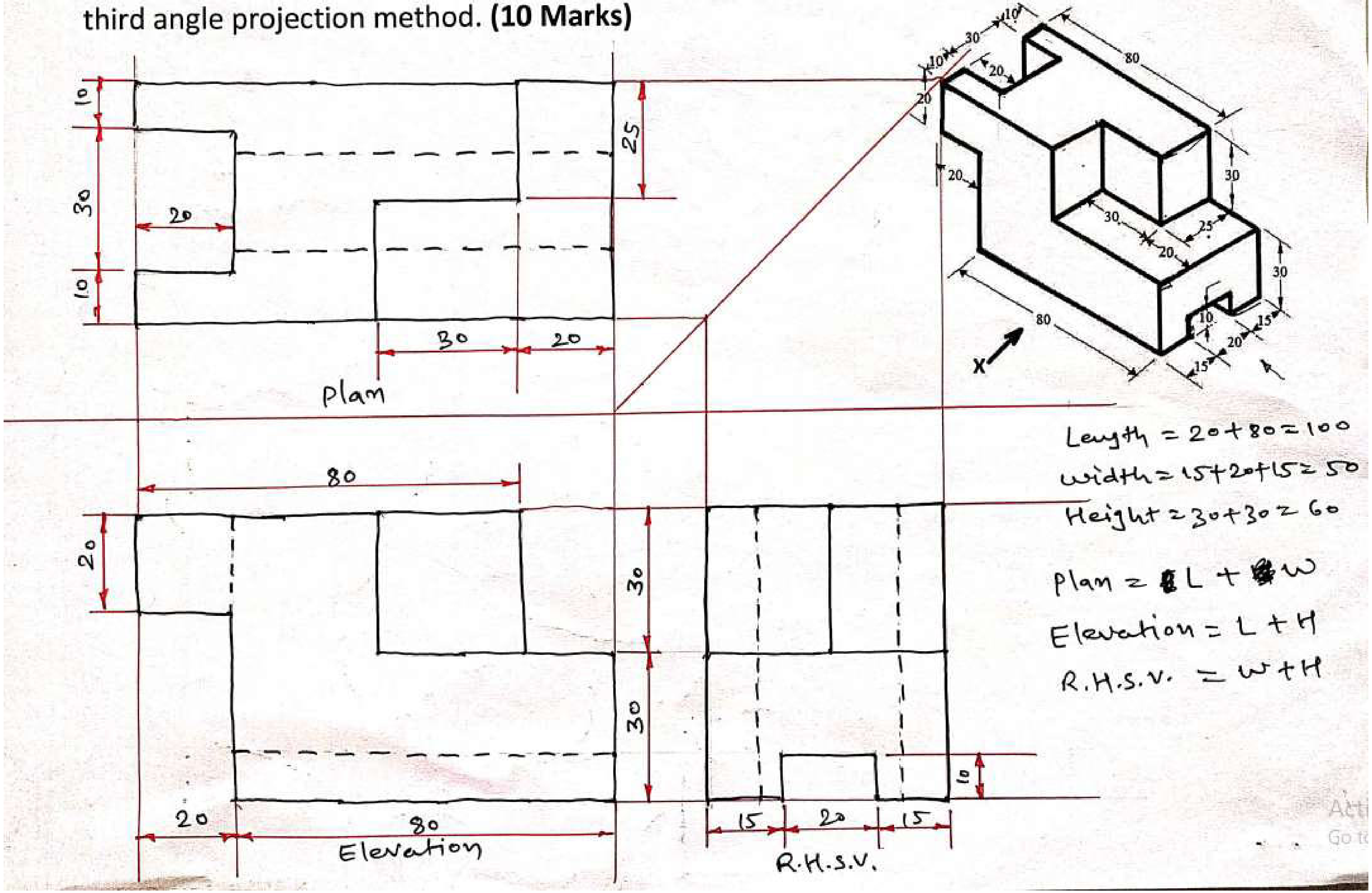
**Advantages of AutoCAD**

1. With the help of CAD highly accurate and precise drawing can be prepared in less time than manually prepared drawings.
2. Drawing instruments and hand skill is not required.
3. Drawings can be easily saved in computer and it can be modified and copied when necessary.
4. We can print the drawings on printer and plotter when required.
5. No scrap of drawings and no paper wastage.
6. Multicolor display makes visualization more effective.
7. We can increase of decrease the size of drawing to definite scale very easily.
8. Computer aided manufacturing is also possible.

**Disadvantages of AutoCAD**

1. Work can be lost because of the sudden breakdown of computers.
2. Work is prone to viruses.
3. Work could be easily “hacked”.
4. Time taking process to know how to operate or run the software.
5. High production or purchasing cost for new systems.
6. Time and cost of training the staff which will work on it.
7. Need of regular updating of software or operating systems.
8. Needs less employment because of CAD/CAM systems.

**Q. 5 (b)** For the figure shown below, draw plan, elevation and right hand side view using third angle projection method. **(10 Marks)**




Length = 20+80 = 100

width = 15+20+15 = 50

Height = 30+30 = 60

Plan = L + W

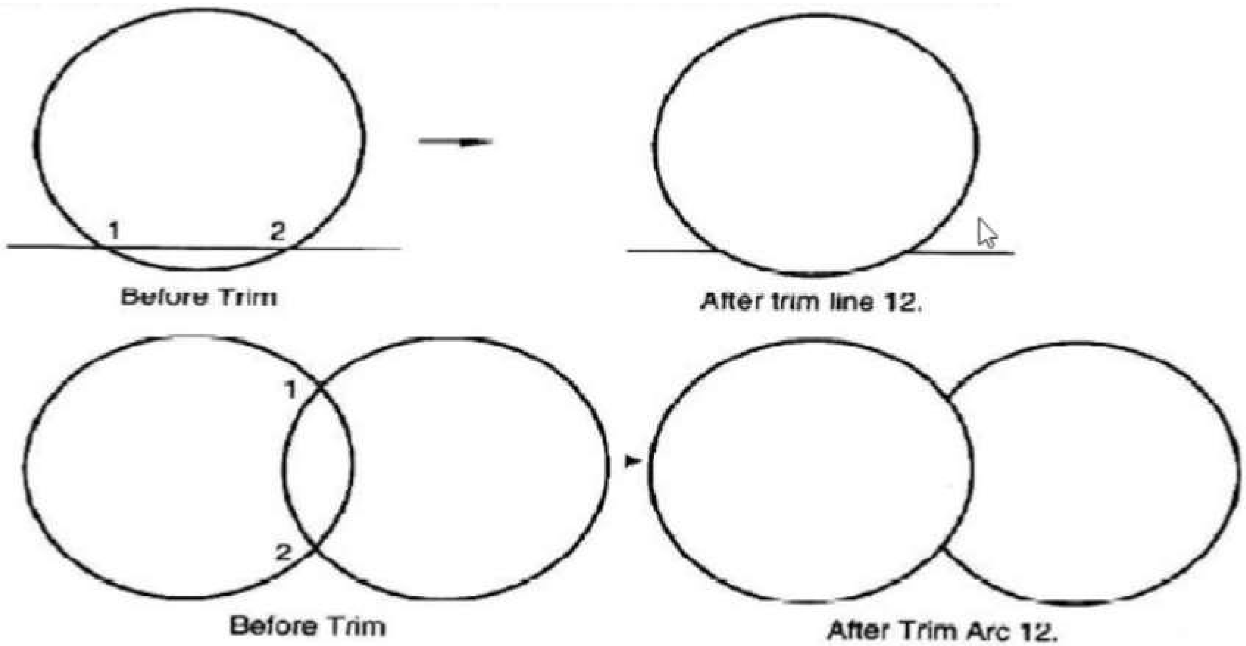
Elevation = L + H

R.H.S.V. = W + H

**Q.5 (a) OR** Explain use of following AUTOCAD commands
1. TRIM 2. ARC 3. MIRROR 4. GRID **( 04 Marks)**

1. TRIM

Trim command is used to erase the portions of selected entries that cross a specified boundary. In fact unwanted entities are removed.



**Q.5 (a) OR** Explain use of following AUTOCAD commands

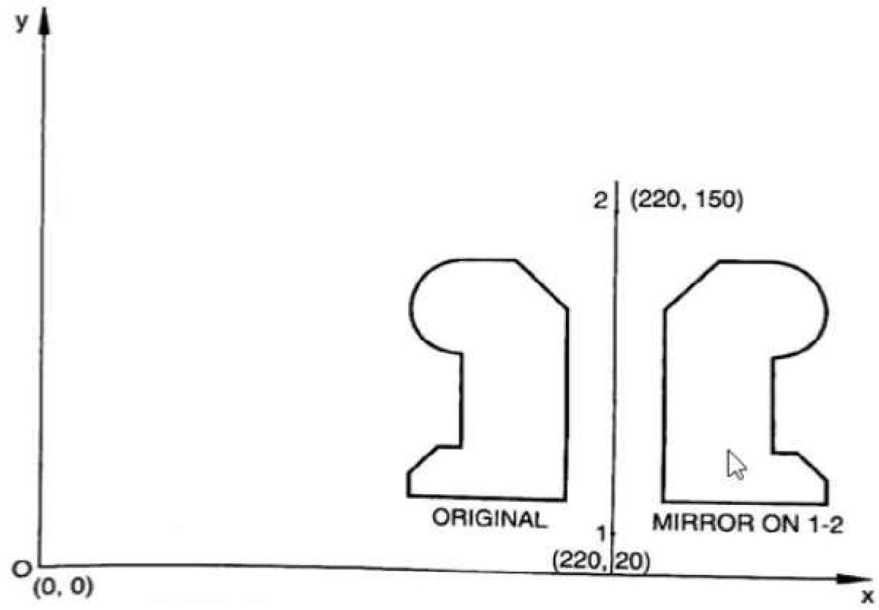
1. TRIM 2. ARC 3. MIRROR 4. GRID **( 04 Marks)**

2. ARC

Arc command is used to draw the arcs as per requirement in the drawing.

3. MIRROR

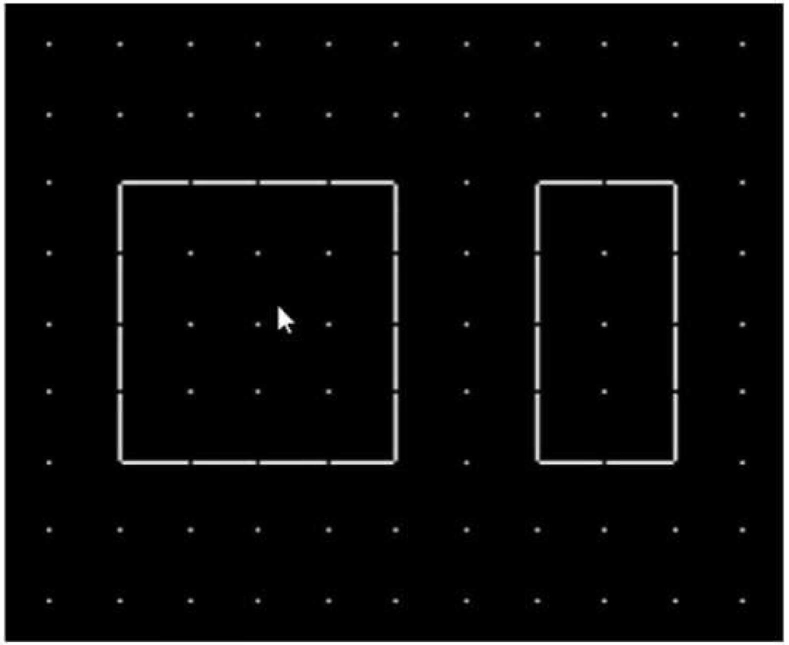
Mirror command is used to obtain mirror image of an object.

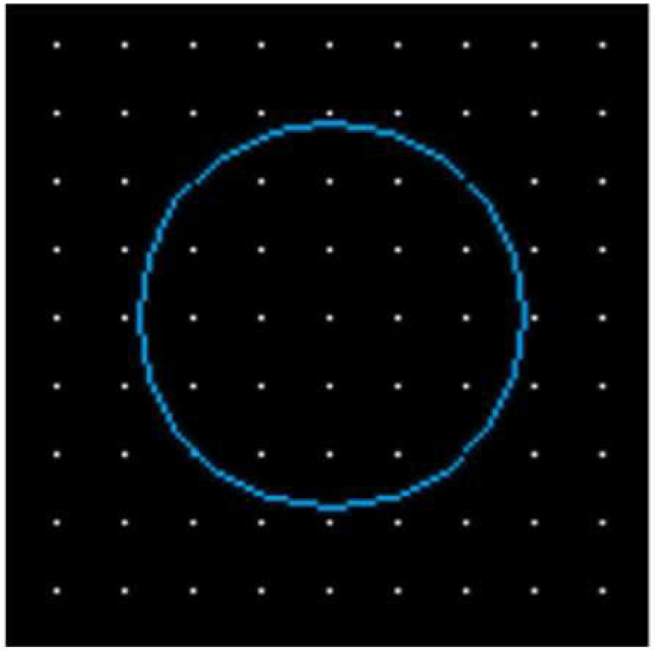


**Q.5 (a) OR** Explain use of following AUTOCAD commands
1. TRIM 2. ARC 3. MIRROR 4. GRID **( 04 Marks)**

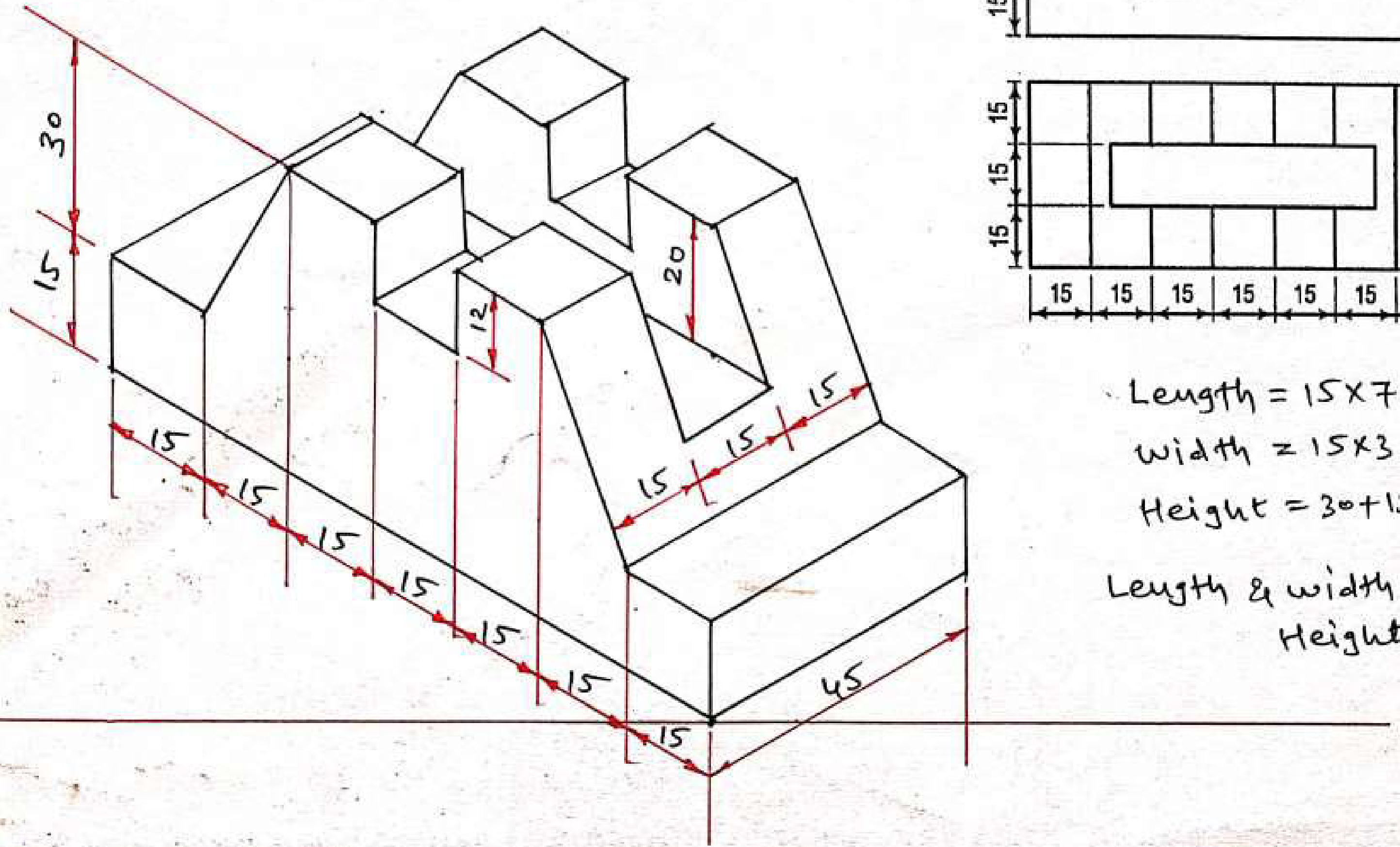
4. GRID

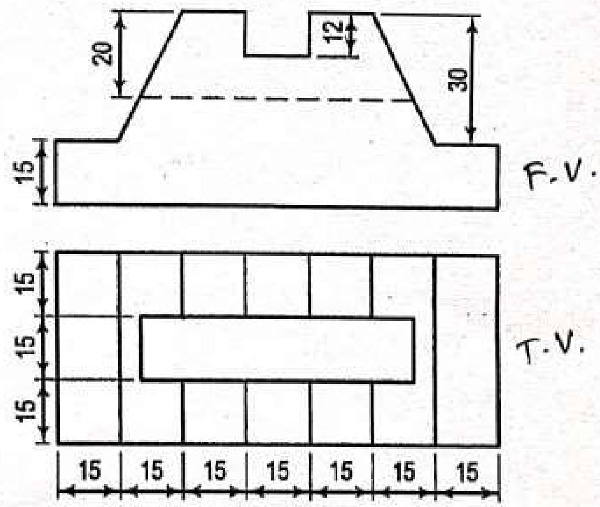
The GRID command is used to display a series of dots or lines in either an equal or unequal horizontal and vertical spacing. The function key to on grid mode is F7. Below is the grid mode on screen of AutoCAD.





**Q. 5 (b) OR** Draw isometric view of the following figure **(10 Marks)**




Length = $15 \times 7 = 105$ mm

width = $15 \times 3 = 45$ mm

Height = $30 + 15 = 45$ mm

Length & width = $30^\circ$

Height = vertical $90^\circ$